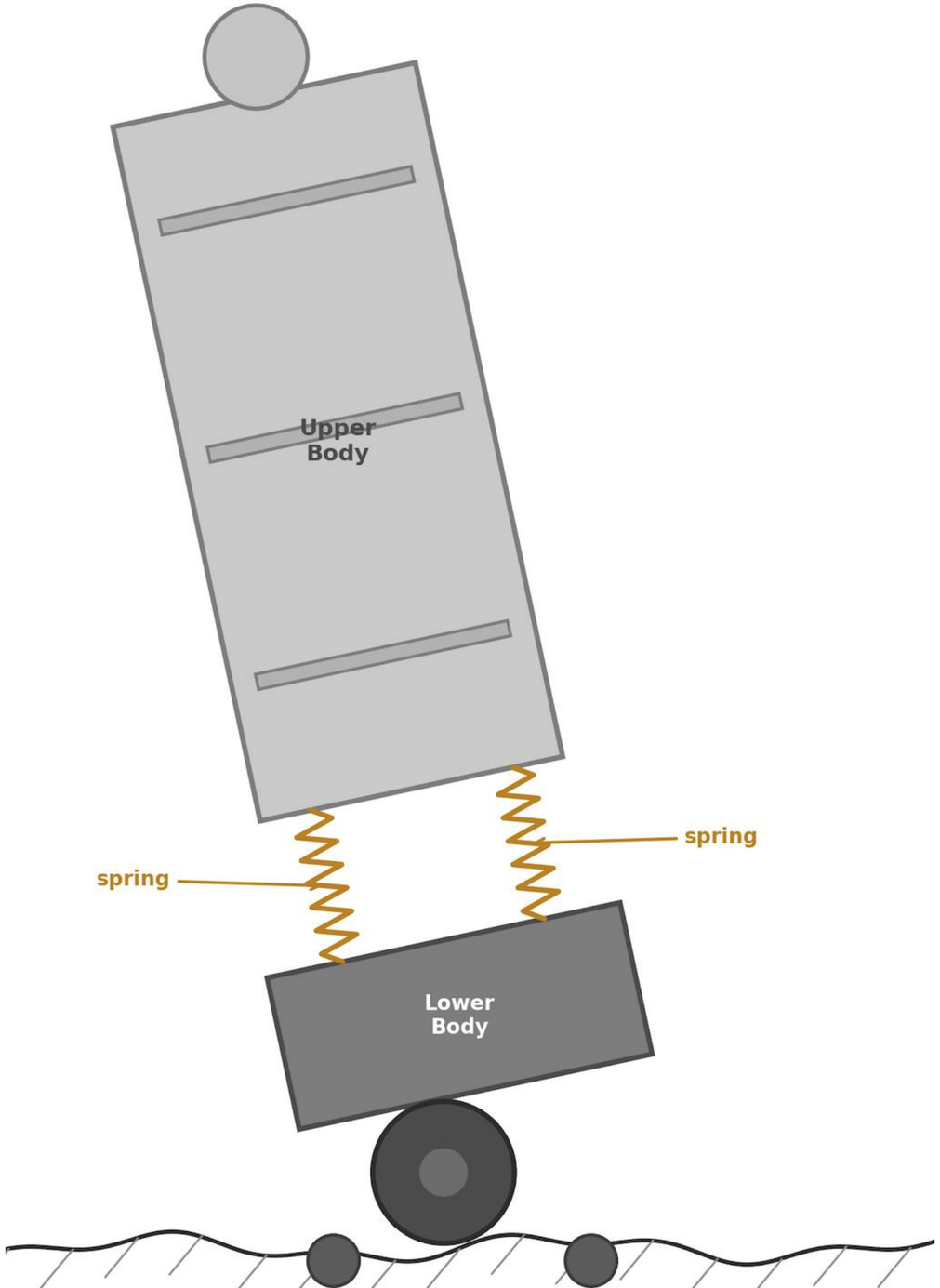
Classical Control vs Reinforcement Learning for Anti-sloshing Control in Food-Serving Robots

MECE 6397 — Learning-Based Systems and Control

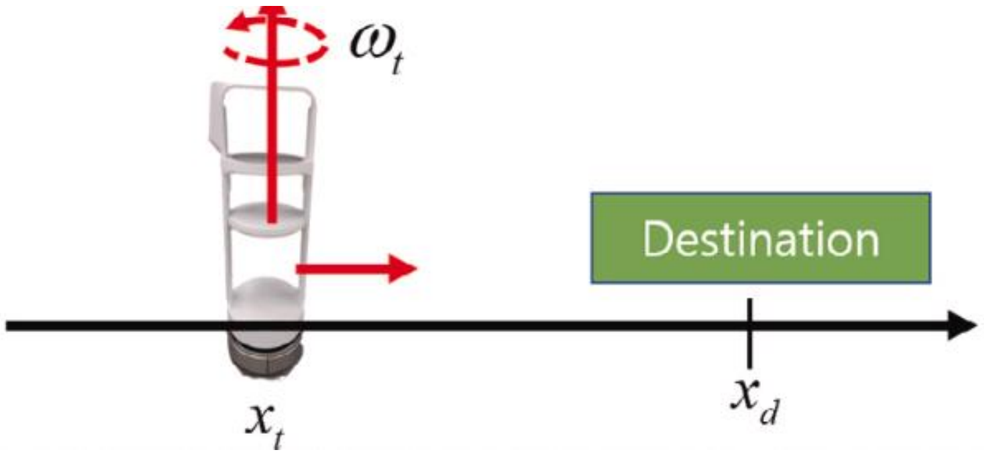
University of Houston | Spring 2025

Motivation

- When carrying **cola, soup, water** — liquids slosh and spill
  - Acceleration / deceleration
  - Turning and curved paths, Uneven floors and human bumps
- Consequences
  - Safety hazards, Customer dissatisfaction
- Current Solution
  - Just move slowly -> Reduces delivery speed and efficiency
  - Stabilizing Tray -> Performs compensatory actions
- Goal
  - Compare classical control vs Reinforcement Learning for anti-sloshing
  - Find a way to control to not spill liquids in the unstable situation


Upper
Body
spring
spring
Lower
Body

Prior Work



- What all previous studies have in common:
  - Every study only considered 1D motion (forward/backward acceleration)
  - The robot moves in a straight line to the destination

- What NO study has considered:
  - Forces during turning (when robot rotates, centripetal force pushes liquid sideways)
  - Vertical forces from floor bumps (changes how heavy the liquid feels)
  - These forces happen constantly in real restaurant environments

Problem Statement

[Objective]

- Compare classical control vs. RL for suppressing liquid sloshing on a serving robot tray

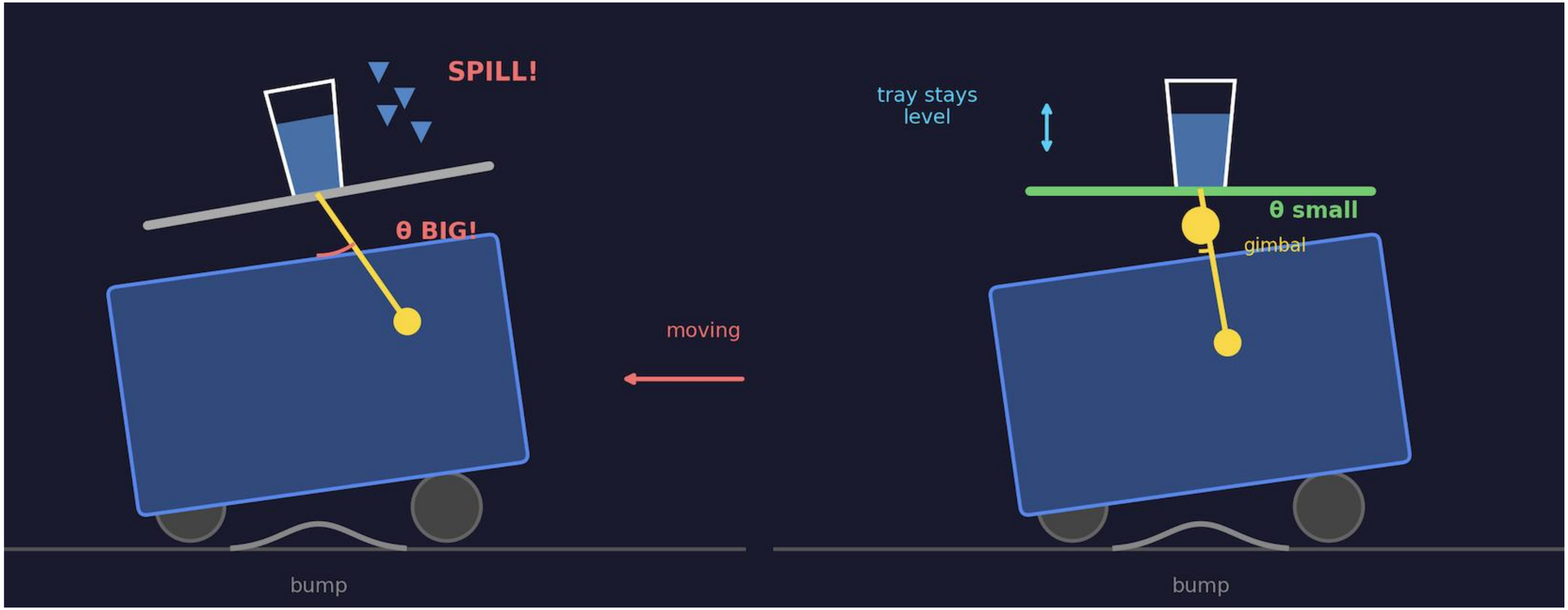
[System]

- A robot tray and glass of water modeled in simulation
- The liquid in a cup behaves like a pendulum swinging inside the cup
- The robot will do 3 types of movement: go straight, turn, and stop suddenly

[Challenges]

- Real liquid in a cup is hard to model
- A full cup behaves differently from a half-full cup, a bowl behaves differently from a tall glass → the physics keep changing
- The controller needs to work in many situations without manually adjusting it each time

Problem Statement



Approach

[Build Simulation]

- On top of the robot, there is a tray that can tilt by itself (like a camera gimbal)
- The tray can tilt forward/backward and left/right using motors
- A pendulum hangs from the tray to represent the liquid in a cup
- The robot experiences 2 types of forces: Horizontal(Acceleration / Break) , Vertical(floor bumps), Sideways(turning)

[Classical Controller]

- Uses PID or LQR to calculate tray tilt angle
- Designed using fixed math equations
- Problem: when floor bumps change the effective gravity, the math becomes wrong

[RL Controller]

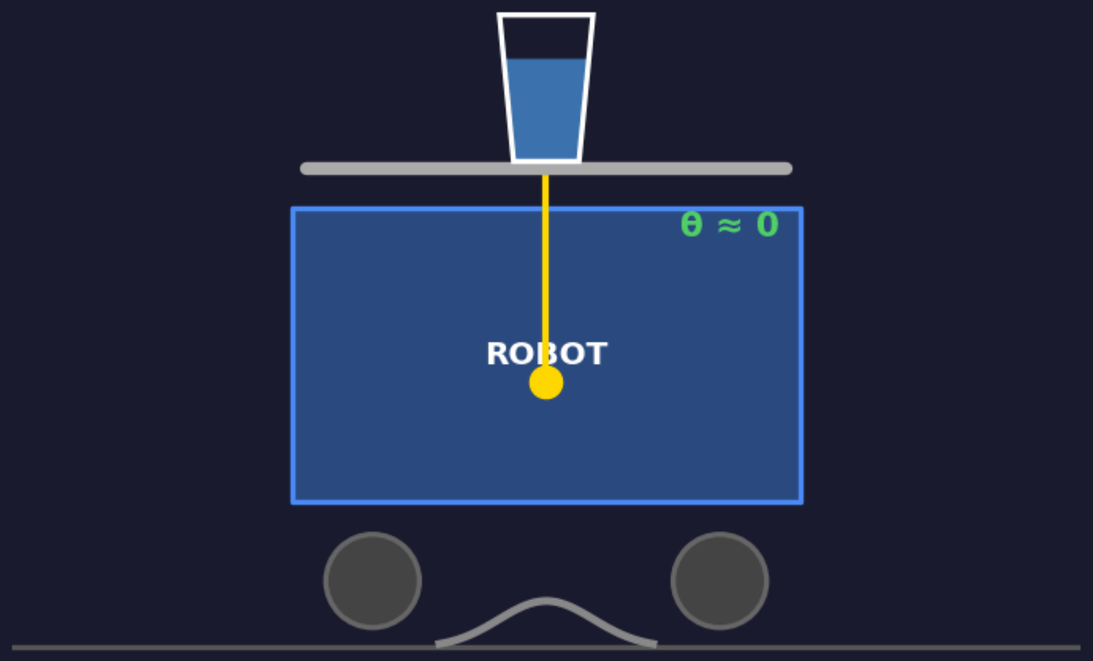
- Train an RL agent to control the tray tilt
- The agent watches: robot speed, tray angle, pendulum swing, vertical acceleration

[Evaluation]

- Situation : straight path, smooth floor // turning paths // turning + floor bumps (vertical disturbances)
- Measure: pendulum swing angle, spill rate, motor energy

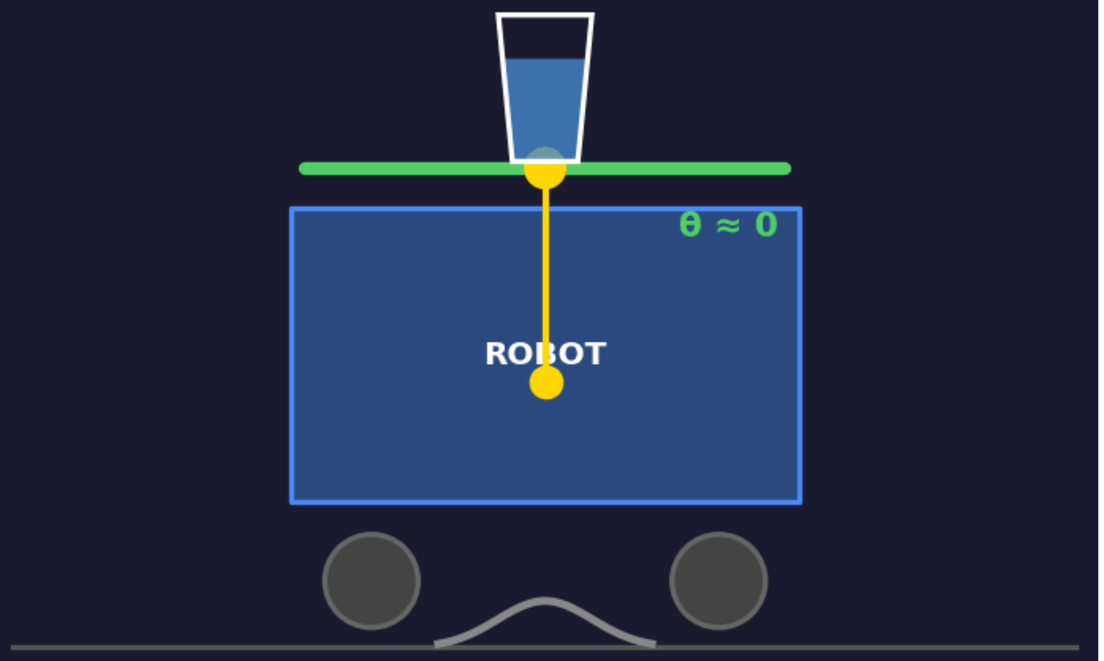
Phase 1: Robot Stationary

Without Gimbal



Everything level

With Gimbal Control



Everything level

THANK YOU